MA1220 - Calculus II – Solution

Date: October 11, 2025 Maximum Marks 45

Time: 120 minutes; 10:00 AM - 12:0 Noon

Q.1 Identify whether each statement is True or False. (No justification required.)

(a) True

(b) False

(c) False

(d) True

(e) False

(f) False

Q.2 (Short Answer) Answer the following.

(a) Let $f(x) = 3x^2 - 12x + 12$. Find all $m > 1$ such that the average value of f on $[1, m]$ equals 3.

(b) Find the area enclosed by the curves $x = \sin y$ and $x = 1 - \cos y$ for $0 \le y \le \frac{\pi}{2}$.

(c) If $f(x, y, z) = \dfrac{xy + yz^2 + xz^2}{x^2 + y^2 + z^4}$ for $(x, y, z) \neq (0, 0, 0)$, does $\lim\limits_{(x,y,z)\to(0,0,0)} f(x, y, z)$ exist?

(d) Determine whether the integral $\displaystyle\int_0^{\pi} \frac{\sin^2 x}{\sqrt{x}}\, dx$ converges or diverges.

(e) Find an equation of the tangent plane to the surface $z = e^{x-y}$ at the point $(2, 2, 1)$.

(f) If $g(2023) = \pi$, assume that g' is continuous, and $\displaystyle\int_{2023}^{2025} g'(x)\, dx = -\frac{\pi}{2}$, find $g(2025)$.

Q.2 (Short Answer) — Solutions.

(a) *Given that the average value for* $f(x) = 3x^2 - 12x + 12$ *on the interval* $[1, m] = 3$. Therefore

$$\frac{1}{m-1}\int_1^m (3x^2 - 12x + 12)\, dx = 3 \implies \frac{m^3 - 6m^2 + 12m - 7}{m-1} = 3 \implies (m-1)^2(m-4) = 0.$$

With $m > 1$, implies that $\boxed{m = 4}$.

(b) *Area enclosed by* $x = \sin y$ *and* $x = 1 - \cos y$, $0 \le y \le \frac{\pi}{2}$. On $[0, \frac{\pi}{2}]$, $\sin y \ge 1 - \cos y$, so

$$A = \int_0^{\pi/2} \big(\sin y - (1 - \cos y)\big)\, dy = \Big[-\cos y + \sin y - y \Big]_0^{\pi/2} = \boxed{2 - \frac{\pi}{2}}.$$

(c) *Limit of* $f(x,y,z) = \dfrac{xy+yz^2+xz^2}{x^2+y^2+z^4}$ *as* $(x,y,z)\to(0,0,0)$. Along $y=x$, $z=0$: $f = \dfrac{x^2}{x^2+x^2} = \frac{1}{2} \to \frac{1}{2}$. Along $y=0$, $z=0$: $f=0$. Since different path give different limits $\Rightarrow$ the limit $\boxed{\text{does not exist}}$.

(d) *The improper integral* $\displaystyle\int_0^\pi \frac{\sin^2 x}{\sqrt{x}}\,dx$ *can be estimated as below*

$$\lim_{t\to 0^+}\int_t^\pi \frac{\sin^2 x}{\sqrt{x}}\,dx \;\le\; \lim_{t\to 0^+}\int_t^\pi \frac{1}{\sqrt{x}}\,dx = \lim_{t\to 0^+}\Big[2\sqrt{x}\Big]_t^\pi = \lim_{t\to 0^+}\big(2\sqrt{\pi}-2\sqrt{t}\big) = 2\sqrt{\pi} < \infty.$$

Hence the improper integral $\boxed{\text{converges}}$.

(e) *Tangent plane to* $z = e^{x-y}$ *at* $(2,2,1)$. Here $f_x = e^{x-y}$, $f_y = -e^{x-y}$, so at $(2,2)$: $f_x = 1$, $f_y = -1$. Thus

$$z - 1 = 1\cdot(x-2) + (-1)\cdot(y-2) = x - y \quad\Rightarrow\quad \boxed{z = 1 + x - y}.$$

(f) Since g' is continuous on $[2023, 2025]$, by the Fundamental Theorem of Calculus,

$$g(2025) - g(2023) = \int_{2023}^{2025} g'(x)\,dx = -\frac{\pi}{2}.$$

Given $g(2023) = \pi$, it follows that $g(2025) = \pi - \frac{\pi}{2} = \boxed{\dfrac{\pi}{2}}$.

Q.3 Answer the following.

(a) Find the area of the region bounded by the graphs of $y = \sin x$ and $y = x\sin(x^2)$ for $0 \le x \le 1$. (4 marks)

(b) Find the volume V obtained by rotating the region enclosed by the curves $y = 2\sqrt{x}$ and $y = x$ about the line $x = -2$. (*Extra note: Do not evaluate the integral; writing the correct integral expression for* V *is sufficient.*)

(c) Find the area of the surface obtained by rotating the curve $x = \sqrt{a^2 - y^2}$ for $0 \le y \le \frac{a}{2}$ about the x-axis.

(d) Let $f : [a,b] \to \mathbb{R}$ be continuous and define

$$g(x) = \int_a^x f(t)\,dt \quad (a \le x \le b).$$

Prove that g is differentiable on (a,b).

Solutions.

(a) *Area between* $y = \sin x$ *and* $y = x\sin(x^2)$ *on* $0 \le x \le 1$. Since $0 \le x^2 \le x \le 1$ and sin is increasing on $[0,1]$, we have $\sin(x^2) \le \sin x$; also $0 \le x \le 1$ gives $x\sin(x^2) \le \sin x$. Hence the

area is

$$A = \int_0^1 \big(\sin x - x\sin(x^2)\big)\,dx = \int_0^1 \sin x\,dx - \int_0^1 x\sin(x^2)\,dx.$$

Now

$$\int_0^1 \sin x\,dx = 1-\cos 1, \qquad \int_0^1 x\sin(x^2)\,dx = \frac{1}{2}\int_0^1 \sin u\,du = \frac{1}{2}(1-\cos 1).$$

Therefore

$$A = \Big(1-\cos 1\Big) - \frac{1}{2}\Big(1-\cos 1\Big) = \boxed{\frac{1}{2}\big(1-\cos 1\big)}.$$

(b) *Volume by rotating the region enclosed by $y = 2\sqrt{x}$ and $y = x$ about $x = -2$.*

Setting $2\sqrt{x} = x$ and squaring both sides gives $4x = x^2$ or $x(x-4) = 0$ so the two curves intersect at $x = 0$ and $x = 4$.

Volume by washers method Rotating the region enclosed by $y = 2\sqrt{x}$ and $y = x$ about $x = -2$ produces a solid whose cross sections are washers with outer radius $R = y - (-2) = y + 2$ and inner radius $r = \frac{y^2}{4} - (-2) = \frac{y^2}{4} + 2$. The volume of the solid of revolution is

$$V = \pi \int_0^4 \big[(y+2)^2 - (\frac{y^2}{4}+2)^2\big]\,dy$$

Volume by cylindrical shells method Using cylindrical shells about the vertical line $x = -2$:

$$V = 2\pi \int_0^4 (\text{radius})(\text{height})\,dx = 2\pi \int_0^4 (x+2)\big(2\sqrt{x} - x\big)\,dx.$$

Note. One can find volume in the previous question by any of the above two methods. Both answers are fine.

(c) *Surface area when rotating $x = \sqrt{a^2 - y^2}$, $0 \le y \le \frac{a}{2}$, about the x-axis (assume $a > 0$).* For rotation about the x-axis using y as parameter,

$$S = 2\pi \int_0^{a/2} y\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\,dy, \quad \text{where} \quad x = \sqrt{a^2 - y^2}.$$

Here

$$\frac{dx}{dy} = \frac{-y}{\sqrt{a^2-y^2}} \quad \Rightarrow \quad \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \frac{a}{\sqrt{a^2-y^2}}.$$

Thus

$$S = 2\pi a \int_0^{a/2} \frac{y}{\sqrt{a^2-y^2}}\,dy.$$

Set $u = a^2 - y^2$, then $du = -2y\,dy$, we get

$$S = 2\pi a\Big[-\sqrt{a^2-y^2}\Big]_0^{a/2} = 2\pi a\left(a - \frac{\sqrt{3}}{2}a\right) = \boxed{\pi a^2\,(2-\sqrt{3})}.$$

(d) We shall show g is differentiable at any arbitrary point on (a, b), say at $x \in (a, b)$. Assume $h > 0$ and $x, x + h \in (a, b)$. Then

$$g(x + h) - g(x) = \int_a^{x+h} f(t)\, dt - \int_a^x f(t)\, dt == \int_x^{x+h} f(t)\, dt.$$

Thus,

$$\frac{g(x + h) - g(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t)\, dt.$$

Since f is continuous on $[x, x + h]$, there exist $u, v \in [x, x + h]$ with $f(u) = m$ (minimum) and $f(v) = M$ (maximum). Using property of definite integrals,

$$mh \leq \int_x^{x+h} f(t)\, dt \leq Mh,$$

i.e.

$$f(u) \leq \frac{1}{h} \int_x^{x+h} f(t)\, dt \leq f(v).$$

Hence

$$f(u) \leq \frac{g(x + h) - g(x)}{h} \leq f(v).$$

Let $h \to 0$. Then $u \to x$ and $v \to x$; by continuity, $f(u), f(v) \to f(x)$. Taking limits,

$$f(x) \leq \lim_{h \to 0} \frac{g(x + h) - g(x)}{h} \leq f(x).$$

By the squeeze theorem,

$$\lim_{h \to 0} \frac{g(x + h) - g(x)}{h} = f(x) \qquad \text{limit exists.}$$

Thus g is differentiable on (a, b).

Q.4 Answer the following.

(a) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(0, 0) = 0$ and

$$f(x, y) = \begin{cases} x \sin \dfrac{1}{x} + y \sin \dfrac{1}{y}, & \text{if } x \neq 0 \text{ and } y \neq 0, \\ x \sin \dfrac{1}{x}, & \text{if } x \neq 0 \text{ and } y = 0, \\ y \sin \dfrac{1}{y}, & \text{if } y \neq 0 \text{ and } x = 0. \end{cases}$$

Show that f is continuous at $(0, 0)$ (give an ε–δ proof). Show that the partial derivatives of f do not exist at $(0, 0)$.

(b) Let $f(x, y) = x^3 + xy^3$. Find all unit directions at the point $(1, 1)$ for which the directional derivative is 0.

(c) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x, y) = x^3$. Find the local maximum and minimum values and the saddle point(s) of f (if they exist).

(d) Find the extreme values of $f(x,y) = 2x + 4y$ subject to the constraint $g(x,y) = x^2 + y^2 - 5 = 0$.

Solutions.

(a) **Checking continuity of f at the origin $(0,0)$.**

Notice that for all $x \neq 0$, $y \neq 0$,

$$|x \sin \tfrac{1}{x}| \leq |x|, \qquad |y \sin \tfrac{1}{y}| \leq |y|.$$

Hence for all (x,y),

$$|f(x,y)| \leq |x| + |y| \leq \sqrt{2}\,\sqrt{x^2+y^2}.$$

Let $\varepsilon > 0$ and choose $\delta = \varepsilon/\sqrt{2}$. If $0 < \sqrt{x^2+y^2} < \delta$, then

$$|f(x,y) - 0| \leq \sqrt{2}\,\sqrt{x^2+y^2} < \sqrt{2}\,\delta = \varepsilon.$$

Therefore $\lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$, so f is continuous at $(0,0)$.

Checking existence of partial derivatives of f at the origin $(0,0)$.

By definition, the partial derivative with respect to x at $(0,0)$ is given by

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \frac{h\sin(1/h) - 0}{h} = \sin\left(\frac{1}{h}\right).$$

The limit $\lim_{h\to 0} \sin(1/h)$ does *not* exist, because $\sin(1/h)$ oscillates between -1 and 1 infinitely often as $h \to 0$. Thus, **partial derivative of f at the origin $(0,0)$ with respect to x does not exist**.

Similarly, the partial derivative with respect to y at $(0,0)$ is given by

$$f_y(0,0) = \lim_{k\to 0} \frac{f(0,k) - f(0,0)}{k} = \sin\left(\frac{1}{k}\right).$$

The limit $\lim_{k\to 0} \sin(1/k)$ does *not* exist. Thus, **partial derivative of f at the origin $(0,0)$ with respect to y does not exist**.

(b) Let $f(x,y) = x^3 + xy^3$. Then

$$\nabla f(x,y) = \left(3x^2 + y^3,\ 3xy^2\right) \quad \Rightarrow \quad \nabla f(1,1) = (4,3).$$

Recall two theorems from the lecture,

- Theorem: If f is differentiable in x and y, then for any unit vector $\mathbf{u} = \langle a, b\rangle$,

$$D_{\mathbf{u}} f(x,y) = f_x(x,y)\,a + f_y(x,y)\,b.$$

- Theorem. If $f_x(x,y)$ and $f_y(x,y)$ exist and are continuous on an open disk $\mathcal{D}$, then f is differentiable on $\mathcal{D}$.

Using these theorems, we can conclude that f is differentiable, and hence for a unit vector $\mathbf{u} = (u_1, u_2)$, the directional derivative at $(1,1)$ is

$$D_{\mathbf{u}} f(1,1) = \nabla f(1,1) \cdot \mathbf{u} = 4u_1 + 3u_2.$$

Thus $D_{\mathbf{u}}f(1,1) = 0$ iff $4u_1 + 3u_2 = 0$ with $u_1^2 + u_2^2 = 1$. From here conclude that

$$\boxed{\mathbf{u} = \left(-\tfrac{3}{5},\, \tfrac{4}{5}\right) \text{ or } \left(\tfrac{3}{5},\, -\tfrac{4}{5}\right)}.$$

(c) Let $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x, y) = x^3$.

Compute first derivatives:

$$f_x = 3x^2, \qquad f_y = 0.$$

Then $f_x = f_y = 0$ implies $x = 0$ (and any y). Hence *every* point on the y-axis is a critical point, i.e., the set of critical points is $\{(0, y) : y \in \mathbb{R}\}$.

Second derivative Test.

$$f_{xx} = 6x, \quad f_{xy} = 0, \quad f_{yy} = 0 \qquad \text{implies that} \qquad D = f_{xx}f_{yy} - f_{xy}^2 = 0.$$

So the second derivative test is inconclusive at *every* critical point.

To check behavious at any point $(0, y_0)$.

First approach $(0, y_0)$ via (h, y_0) for some $h > 0$. Then $f(h, y_0) = h^3 > 0$.

Now approach $(0, y_0)$ via (k, y_0) for some $k < 0$. Then $f(k, y_0) = k^3 < 0$.

Hence there are *no* local maxima or minima; every point on the y-axis is a saddle point:

$$\boxed{\text{No local maxima/minima; saddle points are all } (0, y),\ y \in \mathbb{R}.}$$

(d) Maximise/minimise $f(x, y) = 2x + 4y$ subject to $g(x, y) = x^2 + y^2 - 5 = 0$.

To use method of Lagrange multipliers, set $\nabla f = \lambda \nabla g$, then

$$\nabla f = (2, 4) = \lambda(2x, 2y) \qquad \text{implies that} \qquad (1, 2) = \lambda(x, y).$$

Thus (x, y) is proportional to $(1, 2)$; write $(x, y) = (k, 2k)$ for some k.

The constraint gives $k^2 + (2k)^2 = 5 \Rightarrow 5k^2 = 5 \Rightarrow k = \pm 1$, so the candidates are $(1, 2)$ and $(-1, -2)$.

On evaluating at these points, we get $f(1, 2) = 10$ and $f(-1, -2) = -10$. Thus, we conclude that

$$\boxed{f_{\max} = 10 \text{ at } (1, 2); \qquad f_{\min} = -10 \text{ at } (-1, -2).}$$